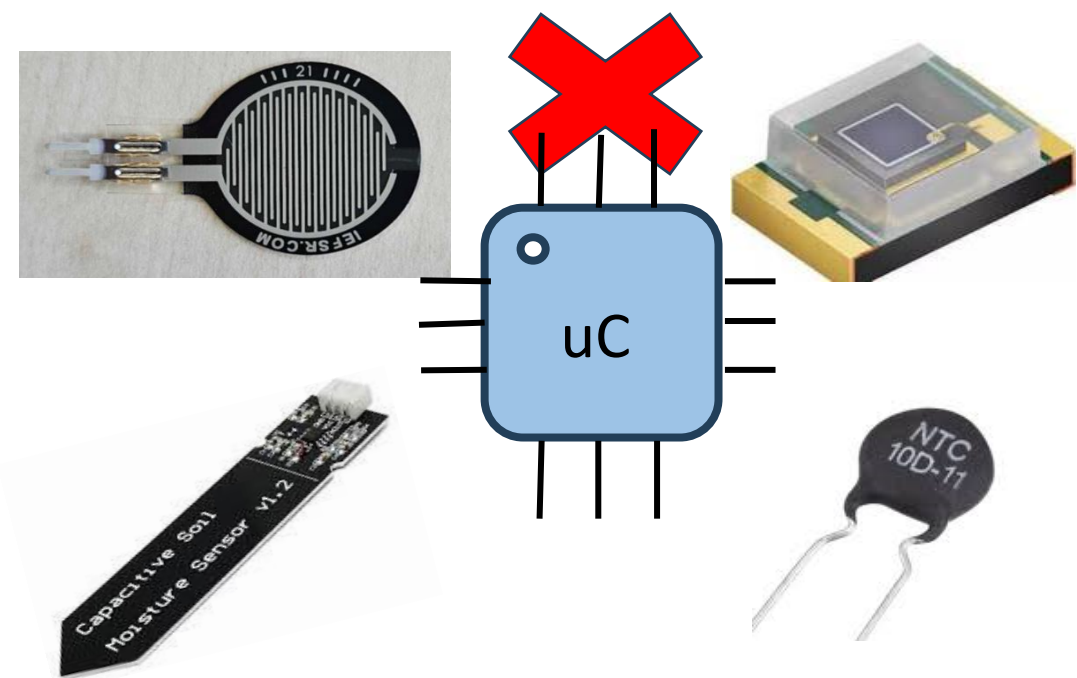




Current Sensor Interfaces

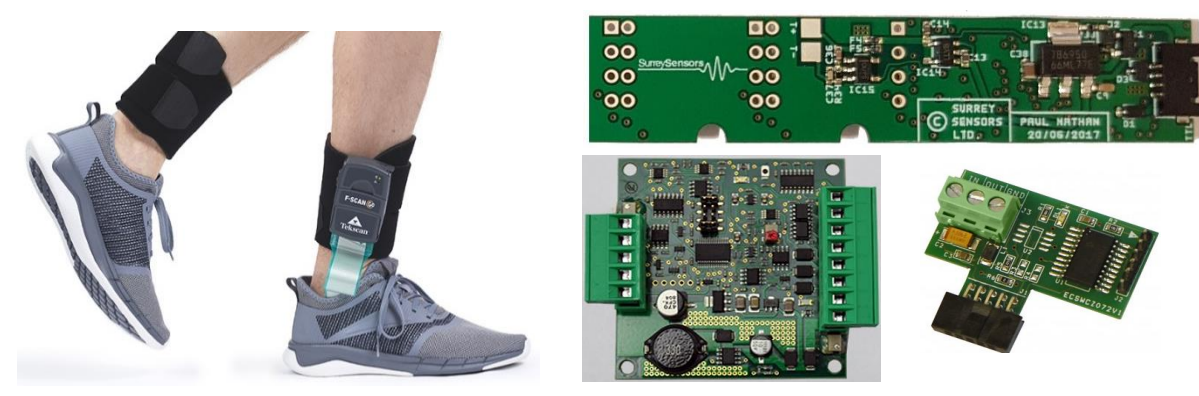
No universal interface for sensing!



Current radio infrastructure does not support battery-free sensing!



Sensor interfaces are bulky, rigid, complex and need batteries

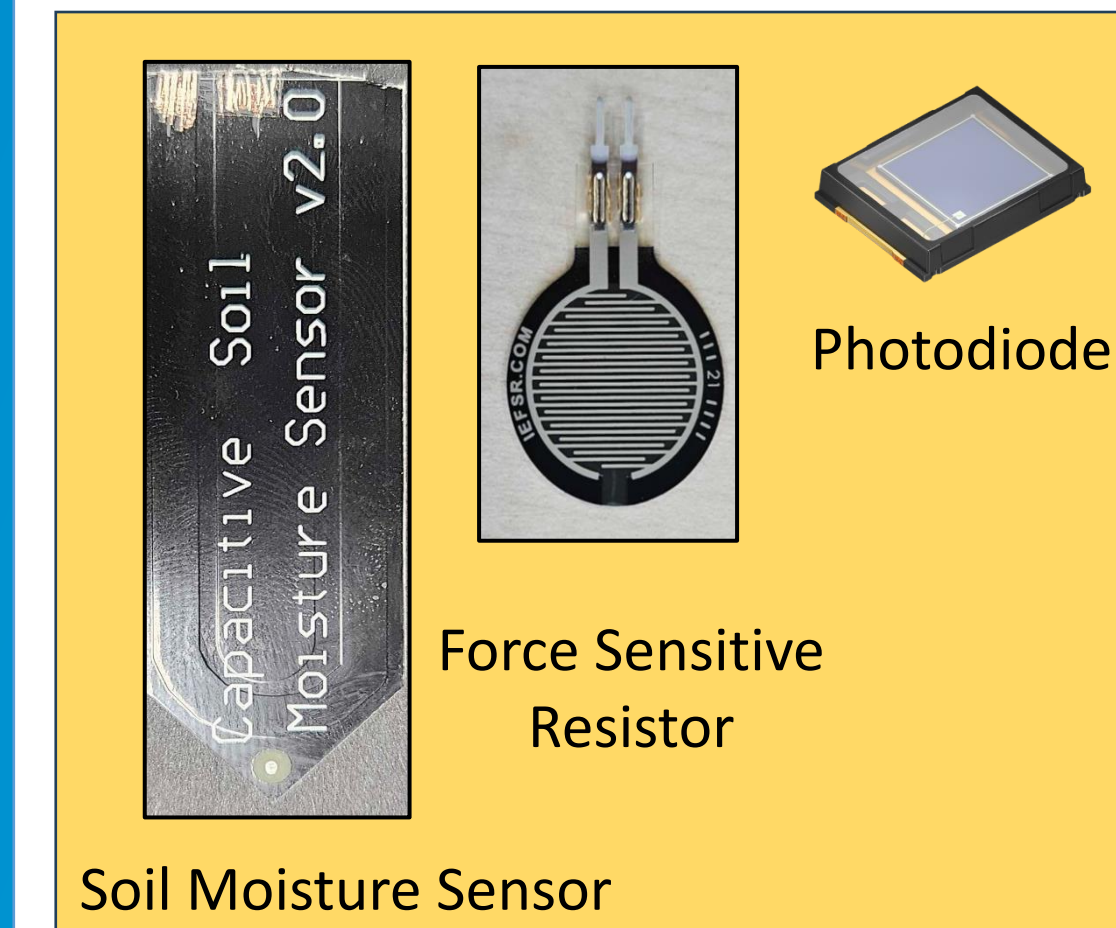


Passive sensing needs SDRs and custom waveforms.



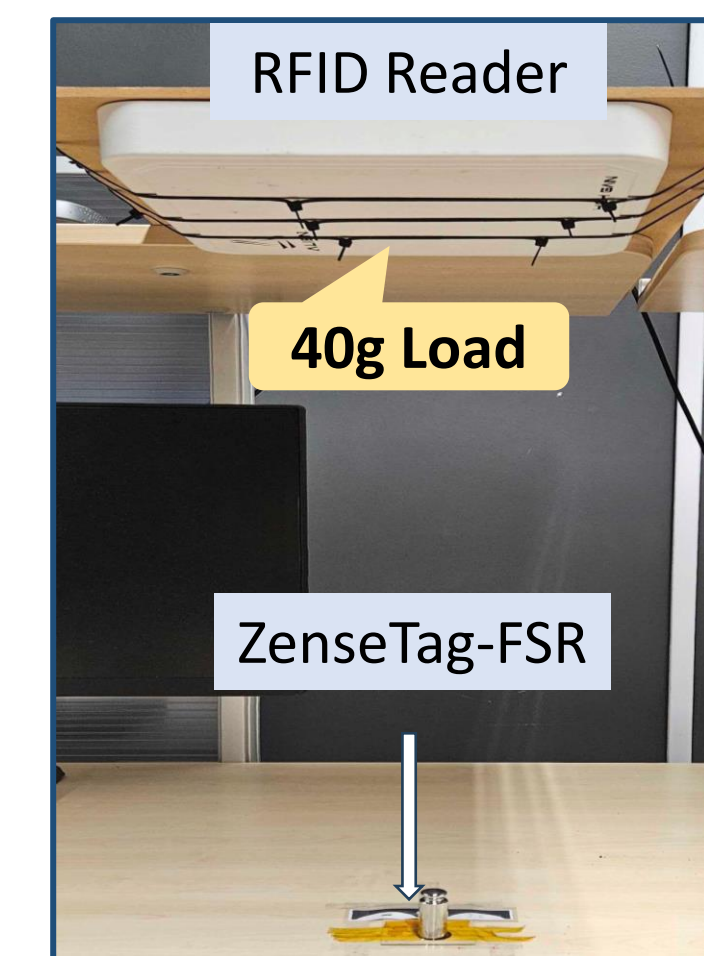
Sensor interfaces are still archaic, bulky, power hungry and non-compliant with existing radios

ZenseTag

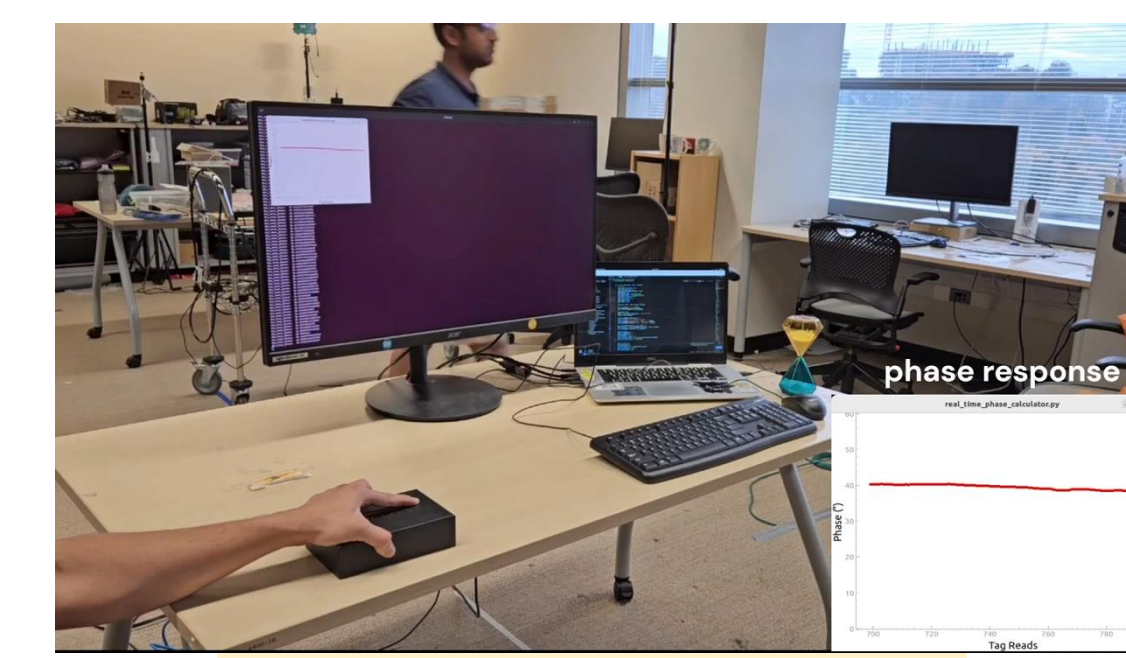


Soil Moisture Sensor

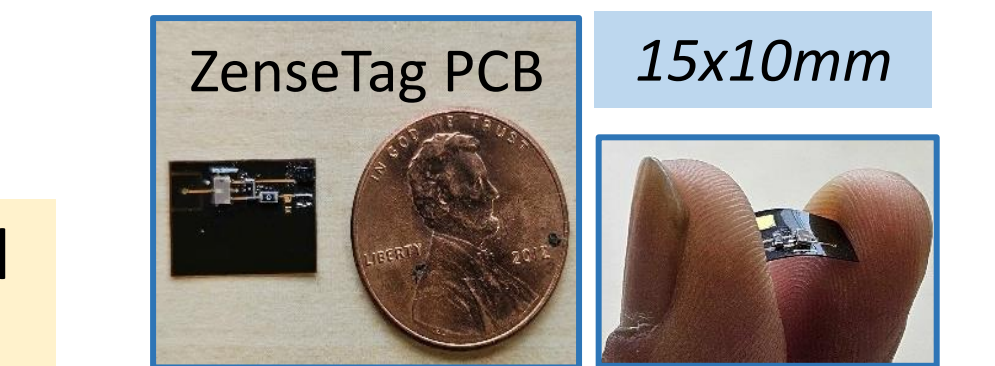
Direct-to-RF COTS Sensor Interface



Batteryless, Commercial RFID Compatible



Robust and Realtime

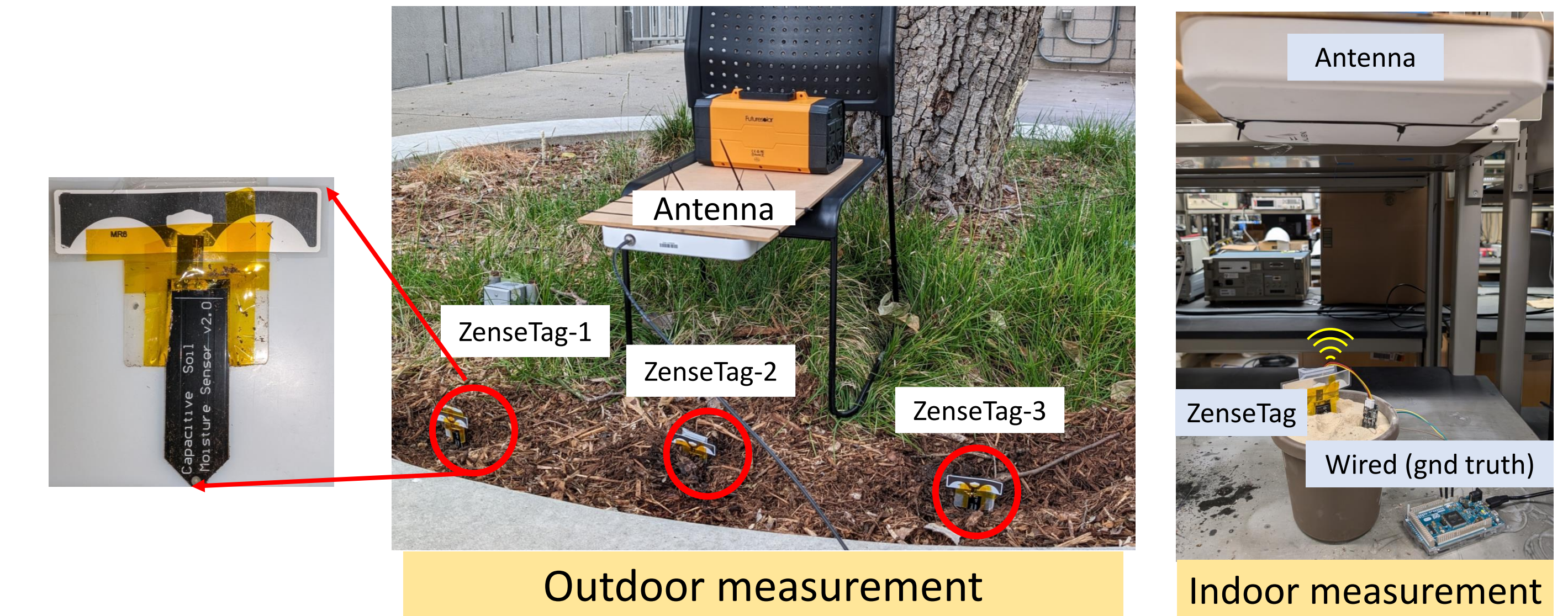


Compact, Flexible form-factor

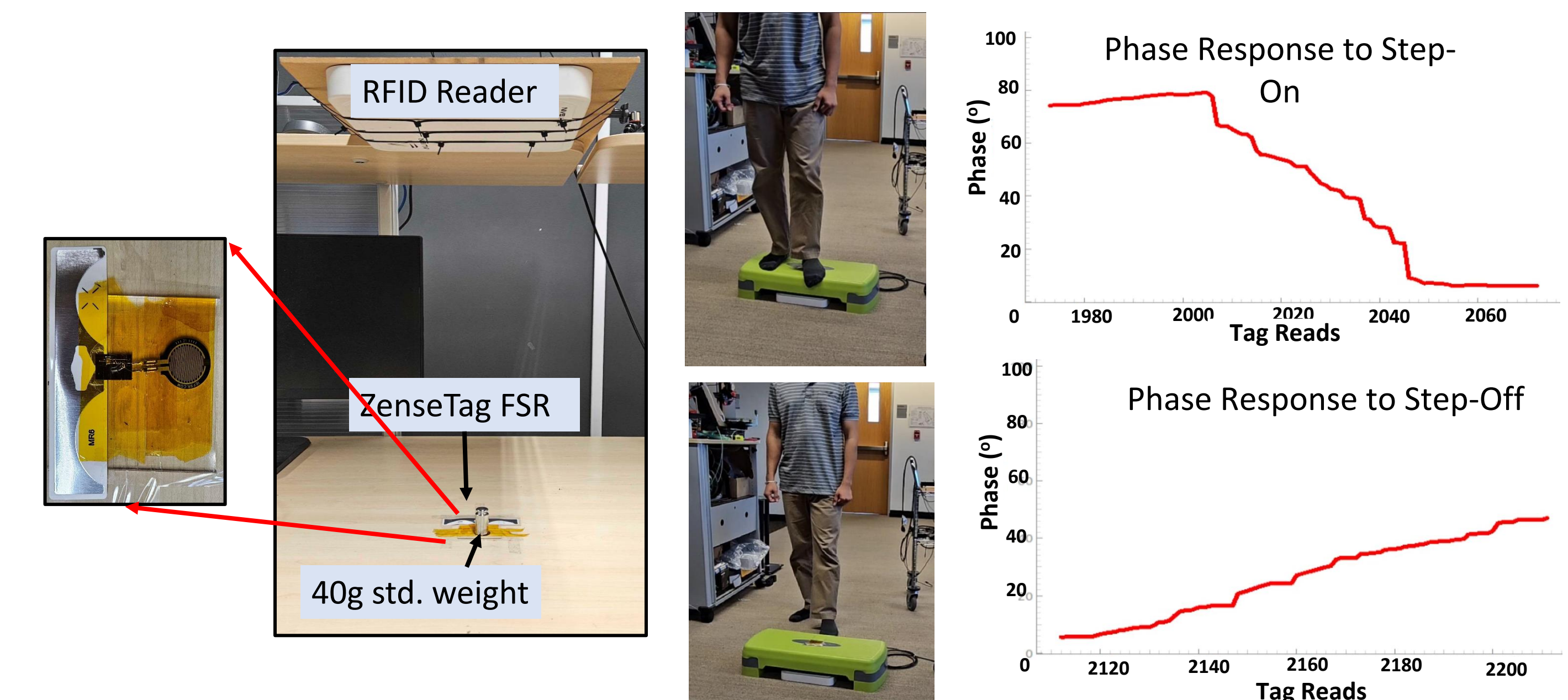
ZenseTag enables a Direct-to-RF interface with commercial antennas using a compact twin-tag-single antenna architecture.

Applications of ZenseTag

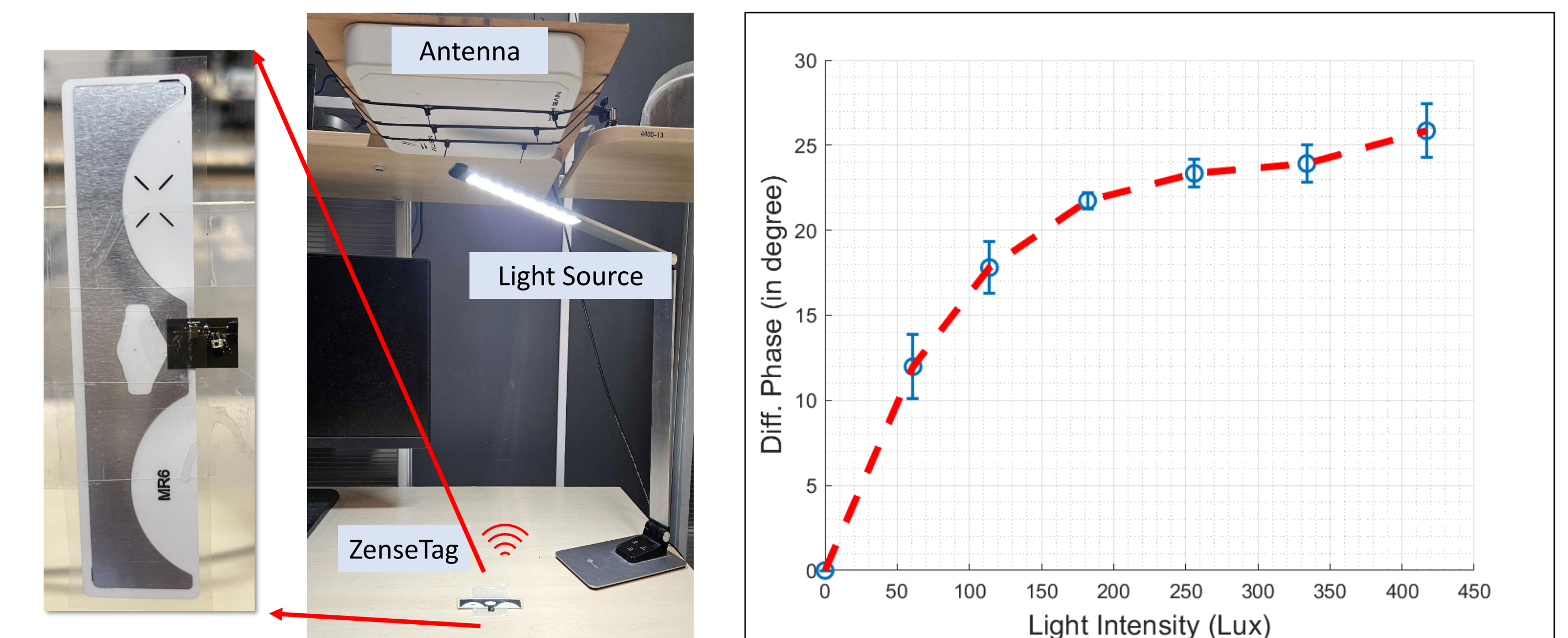
Application 1: Soil Moisture Sensor



Application 2: Contact force sensor

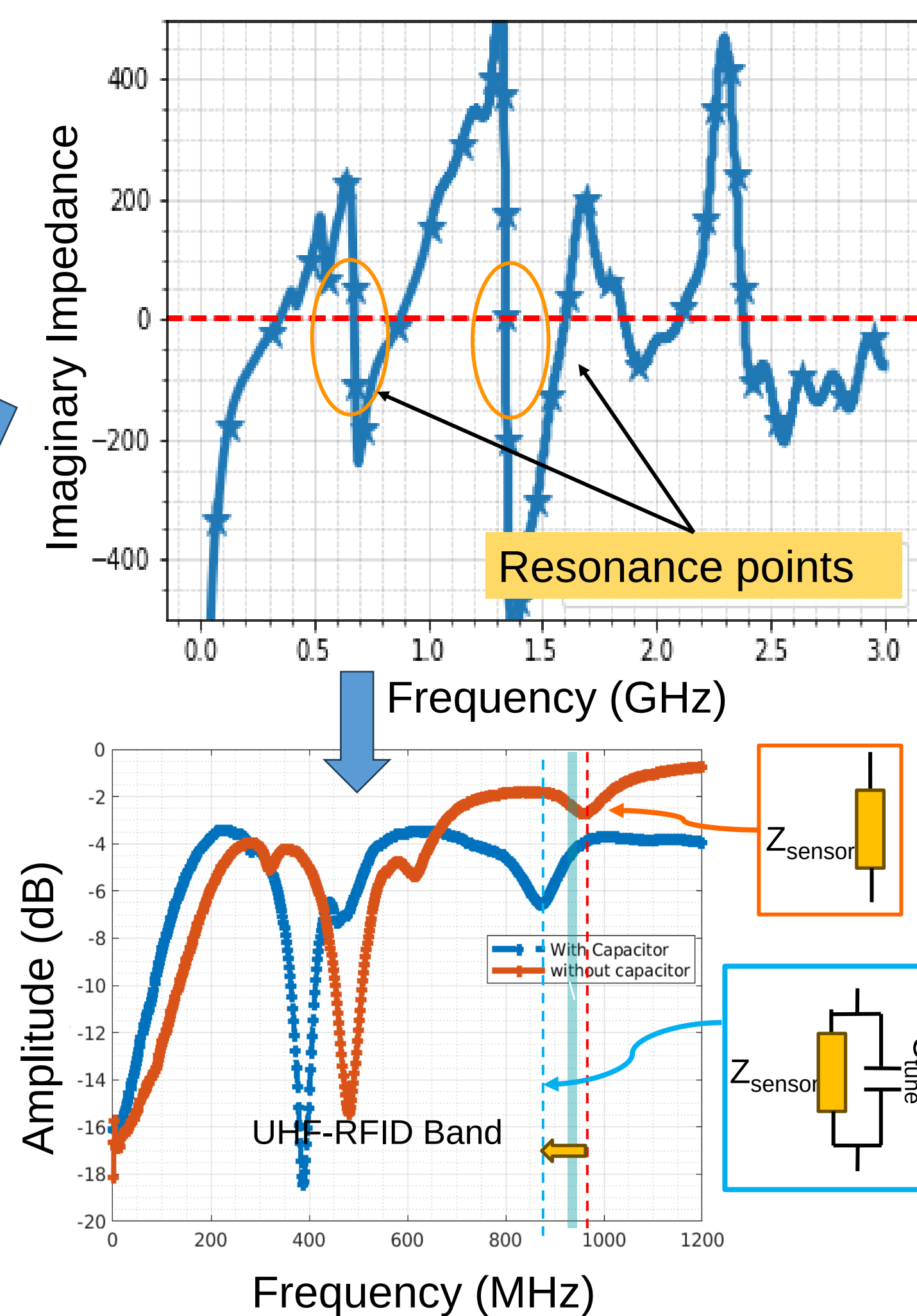
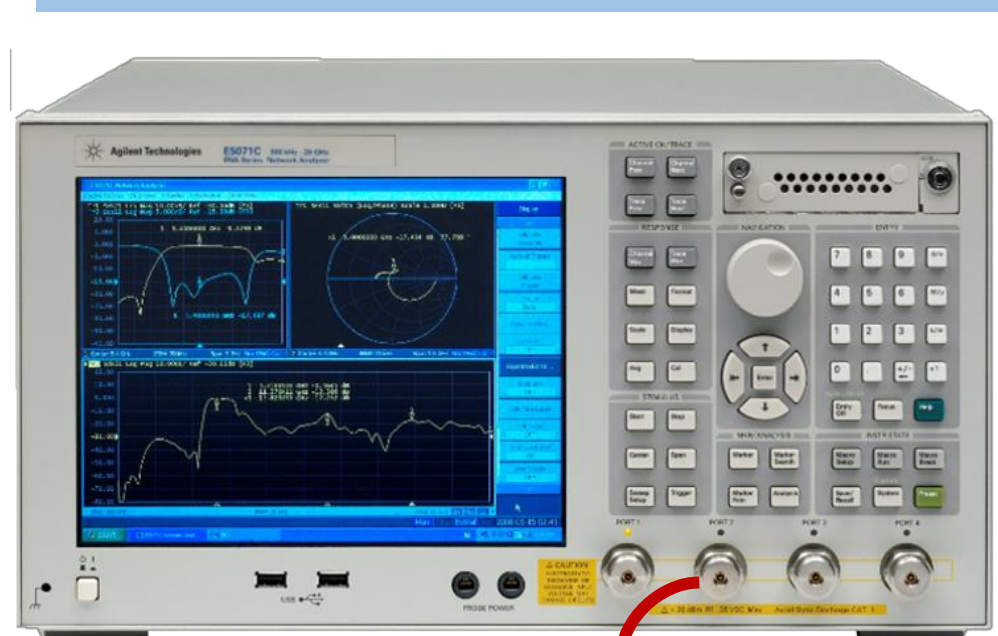


Application 3: Light Sensor



Direct-to-RF Sensor Interface

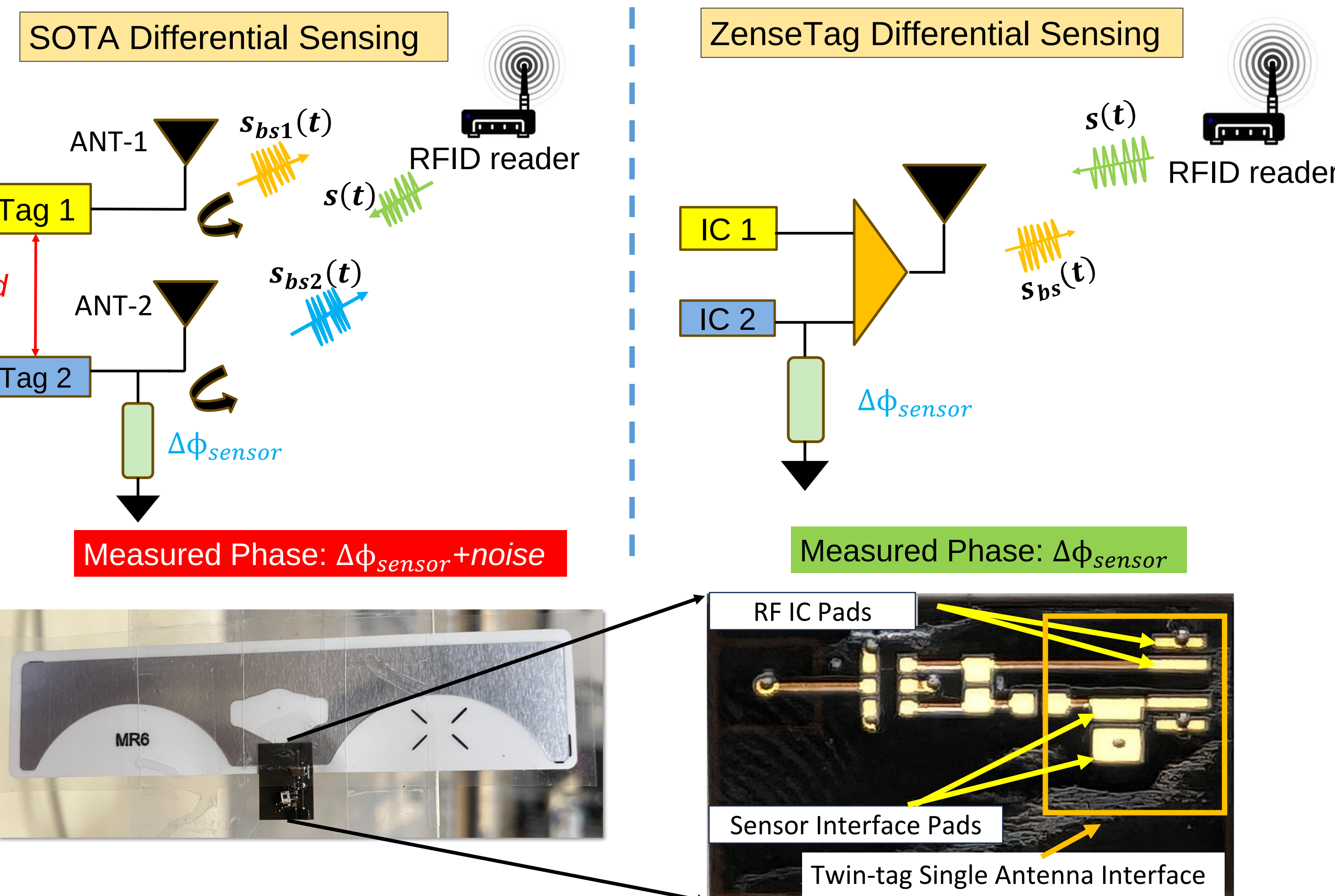
Vector Network Analyzer



Profile Sensor Impedance $Z_{\text{sensor}}(\text{freq})$

ZenseTag enables direct-to-RF-interface by leveraging the resonance of COTS sensors

Twin-Tag-Single Antenna Interface

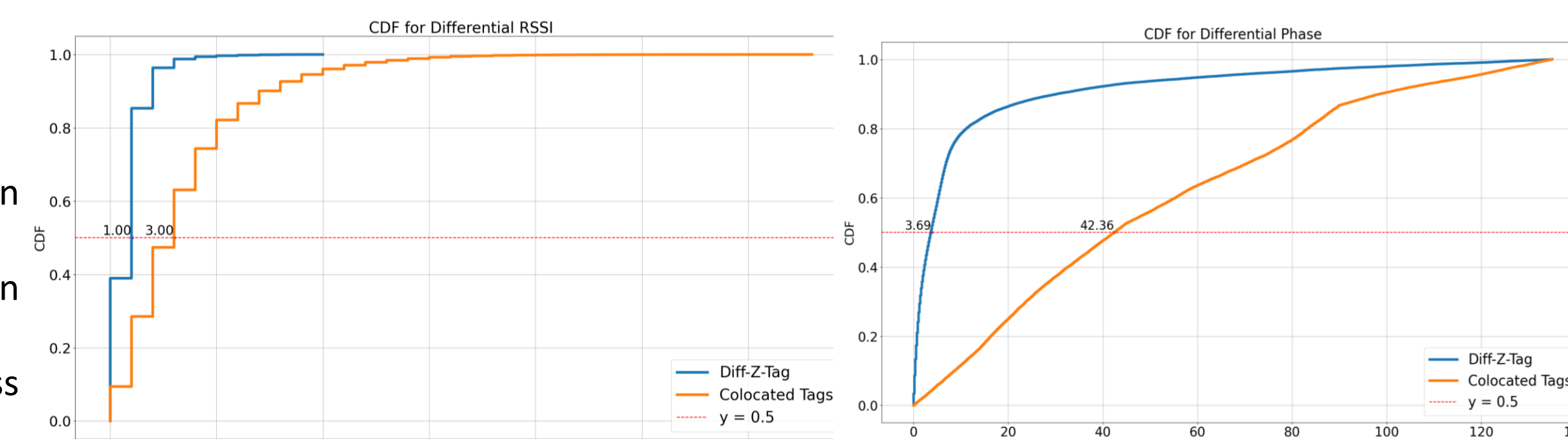


ZenseTag enables robust differential sensing

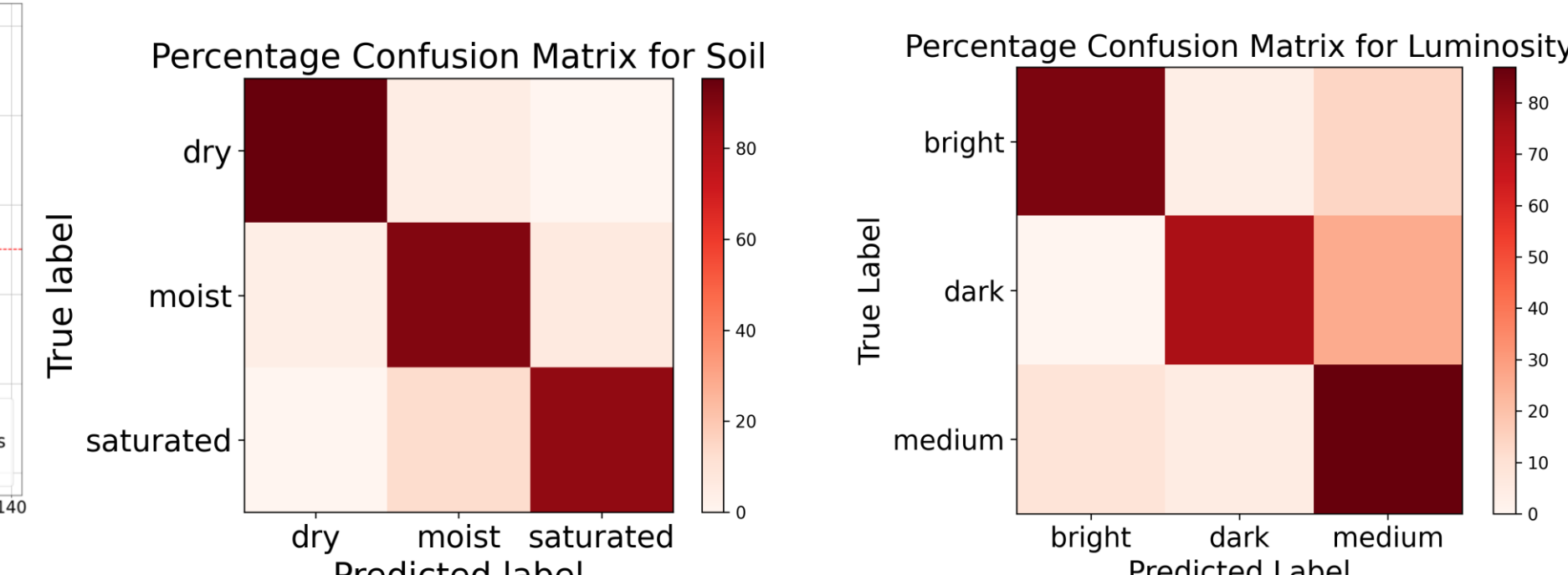
Results

ZenseTag achieves:

- Over 93% accuracy in soil moisture classification (Dry/Moist/Saturated) with three sensors outdoors
- Detects loads as small as 10g (0.1N) and can sense human stepping force in real-time.
- Achieves 85% accuracy in classifying three brightness levels with a photodiode.
- Median phase error below 4°—8x improvement over previous RFID methods.
- Compact, flexible 150mm² PCB



ZenseTag enables **Robust** sensing in multipath



ZenseTag can sense various stimuli accurately

Summary

ZenseTag is a novel RFID-assisted sensing platform that interfaces commercial off-the-shelf (COTS) sensors to affordable, flexible RFID stickers. By utilizing a direct-to-RF interface, it optimally couples sensors with RFID tags. It uses a twin-tag single antenna interface to enable robust readout even in dynamic environments..

